We also tried larger values for alpha, like α=0.995, which are not shown in the table, partly because they lead to poor results: probably the cooling becomes too slow and the algorithm gets closer to random sampling (this suspect is corroborated by the fact that with a smaller value for k, the normalizing Boltzmann constant, the deterioration in the results smoothens).

Table 17.3 Comparing average solution cost found using an ensemble of original methods.

Operation	Mean Cost (α=0.97)	Mean Cost (α=0.98)	Mean Cost (α=0.99)
Adjacent Swaps	1972.502	1937.291	1868.701
Random Swaps	1692.044	1683.563	1668.248
Random Permutation	1816.658	1831.886	1913.416
Ensemble Step	1683.966	1672.494	1660.904

To conclude this section on TSP, listing 17.3 shows a method that implements a random transition from current state to the next one, by combining all three methods discussed so far. An implementation of these methods for library *JsGraphs* can be found on GitHub.